

Overall Findings

The experimental study was conducted to evaluate the effectiveness of different machine learning classification techniques for breast cancer disease prediction using the Breast Cancer Wisconsin Diagnostic dataset. The overall workflow included dataset preprocessing, feature preparation, model implementation, prediction, and performance evaluation using several standard classification metrics.

Three machine learning algorithms—Logistic Regression, Random Forest, and Support Vector Machine (SVM)—were implemented and compared. The models were evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC to obtain a comprehensive understanding of their classification performance.

The experimental results indicate that Logistic Regression and Support Vector Machine (SVM) achieved the best overall performance among the three models. Both models achieved an accuracy of 98.25%, with Precision, Recall, and F1-score of approximately 98.61%. In comparison, Random Forest achieved 95.61% accuracy, 95.89% Precision, 97.22% Recall, and 96.55% F1-score. These results show that Logistic Regression and SVM were more effective in correctly classifying the breast cancer cases in this experiment.

The comparison also demonstrates that model performance can vary depending on the characteristics of the dataset and the preprocessing techniques applied. The high performance of Logistic Regression and SVM suggests that these algorithms were able to effectively identify the patterns and relationships among the diagnostic features of the dataset. Random Forest also demonstrated strong classification performance, particularly in terms of Recall, but its overall performance was slightly lower than that of Logistic Regression and SVM.

Overall, the findings demonstrate that machine learning can be an effective approach for breast cancer classification and prediction. Proper data preprocessing, appropriate model selection, and systematic evaluation play important roles in achieving reliable classification results. Among the models tested in this study, Logistic Regression and SVM were identified as the most suitable models based on the obtained experimental results.